Small Consumer Behavior Change Leading to Market-Wide Depletion During Crisis

Connecting Empirical Evidence with Computational Model

Xucheng Zuo

October 28, 2026

Contents

1 Abstract **2**

2 Introduction **2**

3 Study 1: Rice Supply News and Responses to Shortage Information **2**
3.1 Method . . . 2
3.2 Results . . . 3
3.3 Discussion . . . 5

4 Study 2: A Preregistered Replication with Toilet Paper and Laundry Detergent **5**
4.1 Method . . . 6
4.2 Results . . . 7
4.3 Discussion . . . 8

5 Simulation: Shortage Information on a Network of Close Relationships **9**
5.1 Model . . . 9
5.2 Simulation experiments . . . 13
5.3 Results . . . 14
5.4 Discussion . . . 18

6 General Discussion **19**
6.1 Small individual differences . . . 19
6.2 From individuals to the market . . . 20
6.3 Limitations . . . 20
6.4 Future directions . . . 20
6.5 Conclusion . . . 21

A Appendix **22**
A.1 Behavioral items 22
A.2 Study 1 23
A.3 Study 2 25

1 Abstract

To be drafted

2 Introduction

To be drafted

3 Study 1: Rice Supply News and Responses to Shortage Information

Study 1 examined how consumers respond to shortage information of a staple food. Rice is a staple food in Japanese households, and Japanese consumers experienced a market-wide rice shortage from 2024 to 2025 (i.e., the Reiwa rice crisis). Japan's Ministry of Agriculture, Forestry and Fisheries (MAFF) acknowledged a rice shortage in early summer 2024, which was driven by increasing demand from inbound tourism and a supply shortage caused by extreme weather in 2023 (Ohinata, 2026). Against this backdrop, consumers increased their rice purchases after the Japan Meteorological Agency announced a special advisory on August 8, 2024, warning of an increased risk of a major earthquake (the advisory officially known as the "Nankai Trough Earthquake Extra Information"). This led to severe disruptions in rice distribution across Japan in August and September 2024 (*The Mainichi*, 2024). The study was conducted in January 2026, while people's memory was relatively fresh.

We expected that exposure to news reporting positive or negative prospects for future rice supply would affect consumers' purchase intention. To test this, we manipulated prospects for future rice supply by presenting real news articles (see Appendix) reporting either negative or positive analyses about the future supply. We then measured purchase and information-sharing intentions in two contexts: participants learned of a shortage directly (e.g., during shopping) or indirectly (e.g., from a social media post). The study was exploratory and was not preregistered.

3.1 Method

We recruited Japanese adults through a Japanese survey company, Cross Marketing Inc. (`https://www.cross-m.co.jp/en`), in January 2026 and analyzed 284 of 594 responses (46.8% men; age $M = 45.28$, $SD = 8.87$; exclusions in Appendix). The study was approved by the Ethics Committee of the Graduate School of Humanities and Sociology, The University of Tokyo (UTSP-25047).

Participants were randomly assigned to read two real newspaper articles that were either negative ("Shinmai, Zōsanka no Neagari," 2025; Yasuda, 2025) or positive ("Kome Kakaku ni

Sakiyasukan," 2025; Nakazawa, 2025) about the future supply of rice (n = 165 and 119, respectively). We presented excerpts rather than full articles, selecting sentences about the future supply of rice without changing their wording, so that the two articles in each condition totaled about 540 Japanese characters (542 negative, 536 positive). They then rated their subjective prospects for future rice supply and reported their purchase intention and information-sharing intention in two contexts (see Appendix): finding the rice shelf of their usual supermarket almost empty (*direct observation*) and seeing a social media post with a photo of such a shelf (*indirect observation*). The direct/indirect observation context was a within-participant condition (the order of these two contexts was counterbalanced).

3.1.1 Measures

Subjective Prospects. Participants' subjective prospects for future rice supply were measured with three 5-point scale items: how do you think the supply will change (1 = *very unstable* to 5 = *very stable*), how do you think about it (1 = *very pessimistic* to 5 = *very optimistic*), and how likely do you think a shortage like the Reiwa rice crisis will happen again (1 = *very low* to 5 = *very high*). The first and second item scores were reverse-coded so that higher values indicate more negative prospects, and we aggregated the three item scores confirming their sufficient internal consistency (Cronbach's α = .82).

Purchase Intention. Participants' purchase intention in response to shortage information was measured in each observation context with three 5-point scale items (1 = *definitely not* to 5 = *definitely*; Table 6). In the direct observation context, participants rated whether they would buy the remaining rice at the store when they had rice left at home for a few days, for more than a week, or for more than a month. In the indirect observation context, they rated whether they would go to buy rice right away under the same three levels of household stock.

Information-sharing Intention. Participants' information-sharing intention was measured in each observation context with three 5-point scale items on the same scale (1 = *definitely not* to 5 = *definitely*; Table 6). Two items measured private information-sharing intention: telling housemates or close others, and telling acquaintances or friends, about the shortage observed at the store or on social media. One item measured public information-sharing intention: posting about the store shortage to many unspecified people on social media in the direct observation context, or sharing the post with many unspecified people in the indirect observation context.

3.2 Results

3.2.1 Behavioral variables

Because the behavioral items (i.e., items to measure purchase intentions and information-sharing intentions) were developed for this study, we first conducted an exploratory factor analysis (minimum residual extraction with varimax rotation). Three eigenvalues exceeded 1 (4.96, 2.26 and 1.37; the fourth was 0.92), and the three factors explained 62.7% of the total variance (Table 1; see Appendix A.2.3 for robustness checks). The first factor was related to purchase intention. Regardless of whether participants obtained the shortage information directly or indirectly, and

regardless of how much rice they had at home, six items related to purchase intention highly loaded on the first factor. The second factor was related to private information-sharing intention. Regardless of whether participants obtained the shortage information directly or indirectly, four items about sharing the shortage information with housemates, close others, acquaintances, or friends highly loaded on the second factor. The third factor was related to public information-sharing intention. Regardless of whether participants obtained the shortage information directly or indirectly, two items about sharing the shortage information with many unspecified people on social media highly loaded on the third factor. Following the observed factor structure, we aggregated the six purchase intention item scores, four private information-sharing item scores, and two public information-sharing item scores, respectively.

Table 1: Varimax-rotated loadings of the twelve behavioral items in Study 1 (rice, $N = 284$). Minimum residual extraction; loadings $\geq .40$ in bold. Full item wording is given in Table 6.

Item	Context	Purchase	Private	Public
Buy: a few days left	Direct	**.54**	.29	-.27
Buy: a week left	Direct	**.79**	.23	-.20
Buy: a month left	Direct	**.74**	.10	.14
Go to buy: a few days	Indirect	**.63**	.35	-.05
Go to buy: a week	Indirect	**.80**	.23	.04
Go to buy: a month	Indirect	**.72**	.10	.28
Tell housemates/close	Direct	.22	**.79**	-.04
Tell acquaintances	Direct	.12	**.75**	.26
Tell housemates/close	Indirect	.35	**.68**	.12
Tell acquaintances	Indirect	.26	**.66**	.36
Post to many on SNS	Direct	-.01	.17	**.80**
Share/repost on SNS	Indirect	.02	.15	**.85**
% variance		27.1	20.7	14.9

3.2.2 Effect of the news condition

The manipulation was successful. Participants in the negative prospects condition ($M = 4.05$, $SD = 0.66$) held more negative subjective prospects about the future supply of rice than participants in the positive prospects condition ($M = 3.12$, $SD = 0.76$): $t[231] = 10.75$, $p < .001$, $d = 1.32$. However, the prospects manipulation did not have significant effects on purchase intention (M_{neg} $[SD] = 2.79$ [0.83] vs. M_{pos} $[SD] = 2.73$ [0.78], $t[263] = 0.64$, $p = .521$, $d = 0.08$) and public information-sharing intention (M_{neg} $[SD] = 1.72$ [0.85] vs. M_{pos} $[SD] = 1.70$ [0.75], $t[270] = 0.18$, $p = .853$, $d = 0.02$; see Table 9 in the Appendix). The effect was significant for private information-sharing intention (M_{neg} $[SD] = 2.91$ [0.88] vs. M_{pos} $[SD] = 2.70$ [0.85], $t[258] = 2.02$, $p = .044$, $d = 0.24$).

3.2.3 Subjective prospects and behavioral intentions

As an exploratory analysis, we computed the Pearson's product-moment correlations between subjective prospects and behavioral intention scores across conditions (Table 10 in the Appendix gives the full correlation matrix). The subjective prospects × purchase intention correlation was significant ($r[282] = .18$, 95% CI [.07, .29], $p = .002$), suggesting that participants with more

negative subjective prospects were more willing to buy rice. The subjective prospects × private information-sharing intention correlation was also significant ($r[282] = .18$, 95% CI [.06, .29], $p = .003$), suggesting that participants with more negative subjective prospects were more willing to share the shortage information privately. The subjective prospects × public information-sharing intention correlation was not significant ($r[282] = .05$, 95% CI [-.07, .16], $p = .420$). The correlations did not differ between the news conditions (Fisher's r-to-z test; Table 11 in the Appendix).

3.3 Discussion

The negative prospects news led participants to hold more negative subjective prospects about the supply of rice, but it changed behavioral intentions little: only private information-sharing intention was slightly higher in the negative prospects condition. Across the sample, however, participants with more negative prospects reported higher purchase intention and higher private information-sharing intention. Together, these results suggest that negative subjective prospects about supply are associated with stronger purchase intention and private information-sharing intention when consumers meet shortage information, although the associations were weak.

We think the manipulation did not produce larger behavioral differences because it did not create a contrast between negative and positive prospects. Subjective prospects were negative in the whole sample, and even in the positive prospects condition the mean was above the scale midpoint. The study ran while the 2024 shortage was recent and rice was still expensive, so the positive articles made participants hold less negative prospects rather than positive prospects. The range of subjective prospects was then too narrow for the conditions to differ clearly in behavior. In addition, the comprehension checks excluded more participants in the positive prospects condition, so the two groups may differ in attention.

The behavioral intentions were not organized by the observation context in which the shortage was learned, but into purchase intention, private information-sharing intention, and public information-sharing intention. Study 2 tested this three-variable structure and the associations with subjective prospects in a preregistered study with other commodities.

4 Study 2: A Preregistered Replication with Toilet Paper and Laundry Detergent

In Study 1, both the three behavioral intentions (purchase intention, private information-sharing intention, and public information-sharing intention) and their correlations with subjective prospects were exploratory findings. Study 2 tested both in a preregistered survey (OSF [OSF LINK]) with other commodities, toilet paper and laundry detergent. The survey was conducted in June 2026, during the 2026 Strait of Hormuz oil-supply crisis. Because a real crisis was under way, we did not manipulate subjective prospects; we measured them and related them to the responses consumers reported in this crisis. In May 2026, newspaper reports named both toilet paper and laundry detergent among the daily goods likely to be affected by the oil shock [CITE: news reports, May 2026], but neither had yet run short, so participants responded to a possible shortage rather than to one they were already experiencing.

The preregistered hypotheses were:

- H1. The twelve behavioral items reproduce the three-factor structure of Study 1: purchase intention, private information-sharing intention, and public information-sharing intention.
- H2a. Negative subjective prospects about the supply of the target commodity are positively associated with purchase intention.
- H2b. Negative subjective prospects are positively associated with private information-sharing intention.
- H3. Subjective prospects are not associated with public information-sharing intention.

The preregistration states H2a, H2b and H3 in terms of perceived stability; we report them with the reversed score, subjective prospects (where higher values mean more negative prospects), as in Study 1, so H2a and H2b predict positive associations. The commodity was a between-participants factor included to check generalizability; no difference between commodities was hypothesized.

4.1 Method

We recruited Japanese adults aged 30 to 60, as preregistered, through Cross Marketing Inc. on June 10–11, 2026, and analyzed 536 of 1,028 responses after the preregistered exclusions (53.7% men; age $M = 44.66$, $SD = 8.71$; 257 toilet paper, 279 laundry detergent; exclusions in Appendix A.3.1). The planned sample of 500 gave power of .80 to detect $r = .13$ ($\alpha = .05$, two-tailed), slightly smaller than the smallest relevant correlation in Study 1. The study was approved by the Ethics Committee of the Graduate School of Humanities and Sociology, The University of Tokyo [APPROVAL NO.].

Participants were randomly assigned to toilet paper or laundry detergent and first read a short description of the oil-supply crisis (Supplementary Materials [LINK]). They then rated their subjective prospects about the supply of the commodity and responded to the same direct and indirect observation contexts as in Study 1 (with the order of the two contexts counterbalanced), shown with AI-generated images of the commodity (participants were told that the images were generated). Six attention checks were embedded in the survey (Appendix A.3.1).

4.1.1 Measures

Subjective Prospects. Participants' subjective prospects for the supply of the assigned commodity were measured with three 5-point scale items adapted from Study 1: whether its supply would become unstable, whether the participant could obtain it as usual, and whether it would sell out in stores. Items were coded so that higher values indicate more negative prospects and aggregated (Cronbach's $\alpha = .72$).

Purchase Intention and Information-sharing Intention. Purchase intention (three items per context) and information-sharing intention (two private sharing items and one public sharing item per context) were measured on the same 5-point scale (1 = *definitely not* to 5 = *definitely*) as in Study 1, except that the household stock levels in the purchase items were reworded to suit commodities other than rice (Table 6).

4.2 Results

4.2.1 Behavioral variables

To test H1, we fitted the preregistered confirmatory factor analysis (robust maximum likelihood in lavaan; support criteria: CFI and TLI $\geq .90$, RMSEA and SRMR $\leq .08$) to the twelve behavioral items with three independent factors: purchase intention (six items), private information-sharing intention (four items), and public information-sharing intention (two items). The model did not fit the data ($\chi^2[54] = 1103.70$, $p < .001$, CFI = .715, TLI = .652, RMSEA = .202, SRMR = .214), and the preregistered fallback allowing the factors to correlate also failed the criteria ($\chi^2[51] = 884.59$, $p < .001$, CFI = .768, TLI = .699, RMSEA = .188, SRMR = .089). H1 was therefore not supported by the CFA. Following the next preregistered fallback, we checked the internal consistency of the three item groups; all three were acceptable ($\alpha = .88$ for purchase intention, .87 for private information-sharing intention, and .86 for public information-sharing intention), so we used their item means to test H2a, H2b and H3.

An exploratory factor analysis (minimum residual extraction with varimax rotation, as in Study 1) recovered the three behavioral variables of Study 1 (Table 2). Three eigenvalues exceeded 1 (5.03, 2.40 and 1.21; the fourth was 0.83), and the three factors explained 63.4% of the total variance. Every item loaded highest on its preregistered factor, and the promax solution gave the same pattern (Appendix A.3.2).

Table 2: Varimax-rotated loadings of the twelve behavioral items in Study 2 (toilet paper and laundry detergent, $N = 536$). Minimum residual extraction; loadings $\geq .40$ in bold. Full item wording is given in Table 6.

Item	Context	Purchase	Private	Public
Buy: a few days left	Direct	**.67**	.05	-.21
Buy: half left	Direct	**.81**	.13	-.02
Buy: usual left	Direct	**.76**	.16	.08
Go to buy: a few days	Indirect	**.67**	.23	-.06
Go to buy: half	Indirect	**.75**	.28	.15
Go to buy: usual	Indirect	**.68**	.31	.21
Tell housemates/close	Direct	.25	**.72**	.02
Tell acquaintances	Direct	.13	**.76**	.28
Tell housemates/close	Indirect	.27	**.78**	.08
Tell acquaintances	Indirect	.18	**.73**	.32
Post to many on SNS	Direct	-.02	.21	**.76**
Share/repost on SNS	Indirect	.01	.18	**.90**
% variance		27.7	21.6	14.2

4.2.2 Subjective prospects and behavioral intentions

We tested H2a, H2b and H3 by computing two-tailed Pearson's product-moment correlations between subjective prospects and behavioral intention scores in the pooled sample (Table 3). The subjective prospects × purchase intention correlation was significant ($r[534] = .18$, 95% CI [.09, .26], $p < .001$), supporting H2a, and the subjective prospects × private information-sharing intention correlation was also significant ($r[534] = .20$, 95% CI [.11, .28], $p < .001$), supporting H2b. Contrary to H3, the subjective prospects × public information-sharing intention correlation

was also significant ($r[534] = .19$, 95% CI [.10, .27], $p < .001$), although such sharing stayed rare ($M = 1.67$, $SD = 0.83$). These associations held for both commodities and when concern about the broader oil-supply crisis was controlled (Appendix A.3.3).

4.3 Discussion

Across the two studies, consumers with more negative subjective prospects responded to shortage information with higher purchase intention and higher private information-sharing intention, as hypothesized, with correlations of about .2 in both studies (Table 3). Responses were organized not by the observation context (direct vs. indirect observation) but by whom the behavior was directed at, giving three behavioral intentions: purchase intention, private information-sharing intention, and public information-sharing intention. Both this three-variable structure and the links of subjective prospects with purchase intention and private information-sharing intention replicated across the two studies, whereas the link of subjective prospects with public information-sharing intention was inconsistent ($r = .05$ in Study 1 and .19 in Study 2; Figure 1), and such sharing remained rare in both ($M = 1.71$ in Study 1 and 1.67 in Study 2, on a 1–5 scale).

Table 3: Means, standard deviations and Pearson correlations of subjective prospects and the behavioral intentions in Study 1 (rice, $N = 284$) and Study 2 (toilet paper and laundry detergent, pooled, $N = 536$). $* \ p < .05$, $** \ p < .01$, $*** \ p < .001$.

Variable	*M*	*SD*	1	2	3
Study 1: Rice ($N = 284$)					
1. Subjective prospects	3.66	0.84	–		
2. Purchase intention	2.77	0.81	.18**	–	
3. Private information-sharing intention	2.83	0.87	.18**	.49***	–
4. Public information-sharing intention	1.71	0.81	.05	.05	.30***
Study 2: Toilet paper and detergent ($N = 536$)					
1. Subjective prospects	2.96	0.66	–		
2. Purchase intention	3.44	0.84	.18***	–	
3. Private information-sharing intention	2.94	0.94	.20***	.45***	–
4. Public information-sharing intention	1.67	0.83	.19***	.07	.36***

In Study 2, the preregistered CFA did not fit, even though the EFA recovered the three behavioral variables of Study 1 with every item on the same factor. The preregistered CFA was too strict because it assumed independent factors—whereas purchase intention and private information-sharing intention correlated in both studies—and ignored the shared wording of items repeated across the two observation contexts and the three stock levels.

In both studies, the shortage information was the stimulus: participants saw a context with a nearly empty shelf or an SNS post and reported how they would respond. Subjective prospects, which indicate how strongly a consumer perceives a crisis, were linked to how strongly they responded to this stimulus, but only by a small amount. Whether such small individual changes can produce a shortage in the market as a whole cannot be answered by surveys; the next section addresses it with a simulation in which shortage information triggers purchase and information sharing.

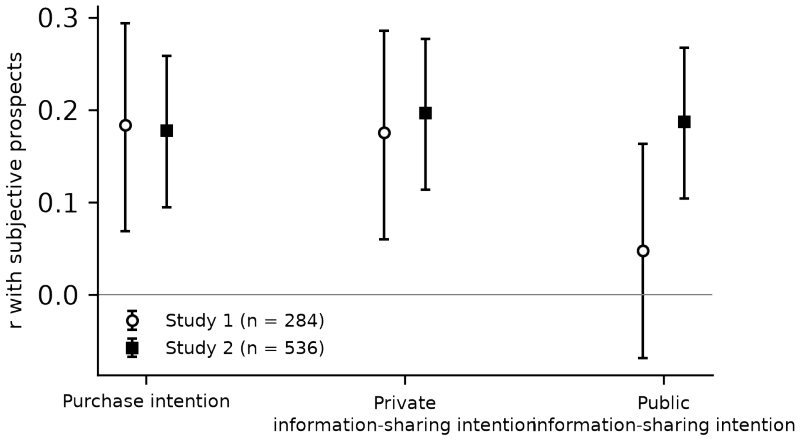


Figure 1: Correlations of subjective prospects with the three behavioral intentions in Study 1 (rice, $N = 284$) and Study 2 (toilet paper and laundry detergent, $N = 536$), with 95% confidence intervals.

5 Simulation: Shortage Information on a Network of Close Relationships

Studies 1 and 2 describe individual intentions. They cannot show what happens when many consumers act on the same shortage at once. The simulation asks whether the small individual differences found in the surveys can add up to a large, fast depletion of stock in a whole market. The model is the network version of our earlier agent-based model [CITE: Zuo & Ohtsubo, 2025]. The earlier version let a consumer see every active sharer within a spatial radius. The surveys suggest a different channel: people mostly tell family, partners and friends, and these close others do not have to live nearby. The network version therefore replaces the spatial radius with a small number of close relationships on a small-world network [CITE: Watts & Strogatz, 1998].

5.1 Model

5.1.1 Overview

The model contains consumers, retailers and one commodity. Consumers shop when they are exposed to shortage information, retailers run out, and consumers who fail to buy may tell their close contacts. Figure 2 shows this loop. The loop stays quiet if too few consumers hear the information, if those who hear do not react, or if those who witness a shortage do not tell anyone. The experiments below therefore change these links one at a time and then together.

5.1.2 World and consumers

The market contains 2,500 consumers and 25 districts. Districts sit on a ring, and consumer identifiers run around the same ring, so consumer c lives in district $\lfloor cD/N \rfloor$, where D is the number of districts and N the number of consumers. Each district has 100 residents and one retailer. Stock is set by an abundance of .5: each retailer holds 50 units, and the whole market holds 1,250 units for 2,500 consumers. Stock is not replenished, so the model covers a short

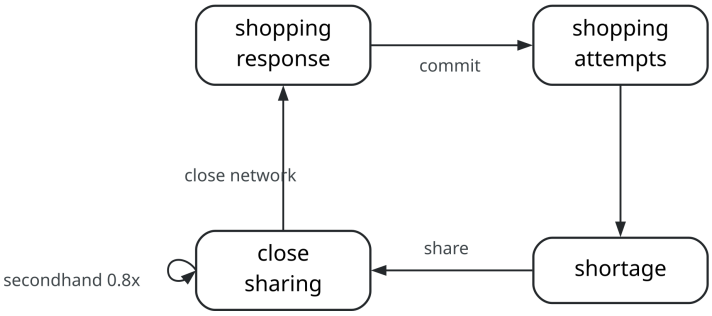


Figure 2: The feedback loop in the model. The shopping response turns received information into shopping attempts, attempts empty districts, failed attempts produce shortage information, and close sharing passes it through the close network. Heard information can be forwarded once at 80% of the close-sharing probability.

period in which the only question is how fast the fixed stock runs out. District 0 starts with no stock. This seeded shortage is the only source of shortage information at the start of every run (Figure 3).

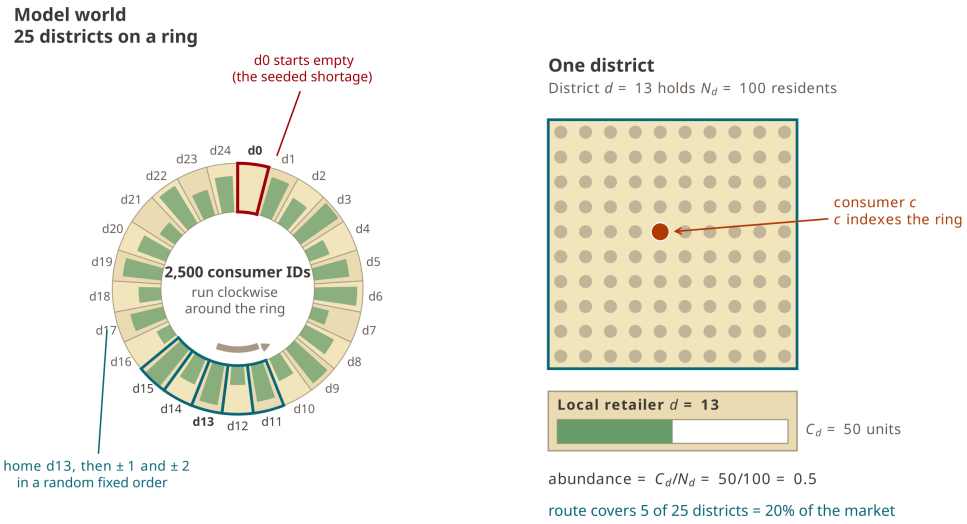


Figure 3: The 25 districts on a ring. District 0 starts empty. A consumer's shopping route is fixed at setup: the home district first, then the two districts on each side in random order.

A consumer makes one shopping commitment and buys at most one unit. Once committed, the consumer goes to the home district first. If the district has run out, the consumer tries the next district on a fixed route of five districts (home, then two on each side, in an order drawn at setup). A consumer who fails at all five gives up. When a district has fewer units than shoppers in a step, the remaining units are allocated at random among that step's shoppers.

5.1.3 Close relationships and rewiring

Information moves only through close relationships. The network is a Watts–Strogatz graph built on the consumer ring. Each consumer starts with 10 close relationships, the five consumers on each side. Because a 10-person neighborhood sits well inside a 100-resident district, 97% of these links connect consumers of the same district, and information on the unrewired ring stays almost

entirely local. Each link is then rewired with the *rewiring probability*: one endpoint is replaced by a randomly chosen consumer (Figure 4). Rewiring keeps the total of 12,500 links fixed while individual degrees change. The rewiring probability replaces the information radius of the earlier model as the parameter for how far information can reach. The network is generated once per run from its own random stream and is then fixed.

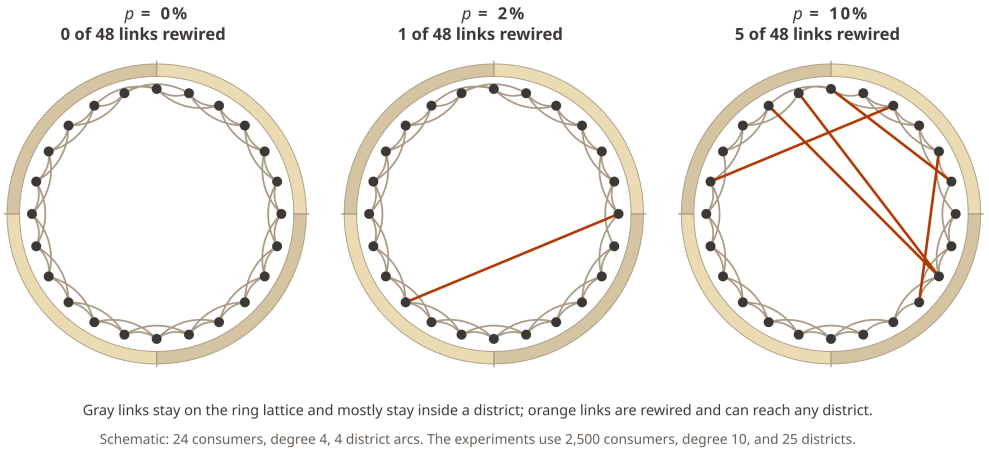


Figure 4: Watts–Strogatz rewiring on the consumer ring. Gray links belong to the original lattice and mostly stay inside a district. Orange links have been rewired and can land anywhere on the ring.

5.1.4 Sharing

A consumer who fails to buy in an empty district holds *firsthand* shortage information. With the close-sharing probability p_{close}, this consumer tells all current contacts at once. Sharing is undirected, so either end of a link can hear the other, and a sharer stays visible to all contacts for 10 steps. A consumer who hears about the shortage for the first time gets one chance to forward it as *secondhand* information, with probability $0.8 \times p_{close}$. There is no separate secondhand trait. This follows the survey results: in both studies, firsthand close sharing was higher than secondhand close sharing, but the two were strongly correlated (see *Firsthand and secondhand sharing* in the Results of both studies). The ratio of .8 sets the shape of the assumption, not a calibrated value, because intentions on a 1–5 scale are not probabilities. Public SNS posting is not modeled.

5.1.5 Shopping response

The number of contacts currently sharing, $I_{i,t}$, sets the shopping probability through a sigmoid response:

$$P_{i,t}(\text{shop}) = P_0 + \frac{L - P_0}{1 + e^{-G(I_{i,t} - M)}} \quad \text{if } I_{i,t} > 0,$$

and $P_{i,t}(\text{shop}) = P_0$ if no contact is sharing. Here $P_0 = .01$ is the baseline shopping probability per step, $L = .10$ is the maximum, $G = 20$ is the steepness and M is the *reaction midpoint*. With this steepness the response is close to a step, so the midpoint works almost as a threshold count of contacts sharing at the same time. Against a mean degree of 10, a midpoint of 10

means a consumer needs the whole neighborhood to share at once, while a midpoint below 1 means one sharing contact is enough (Figure 5).

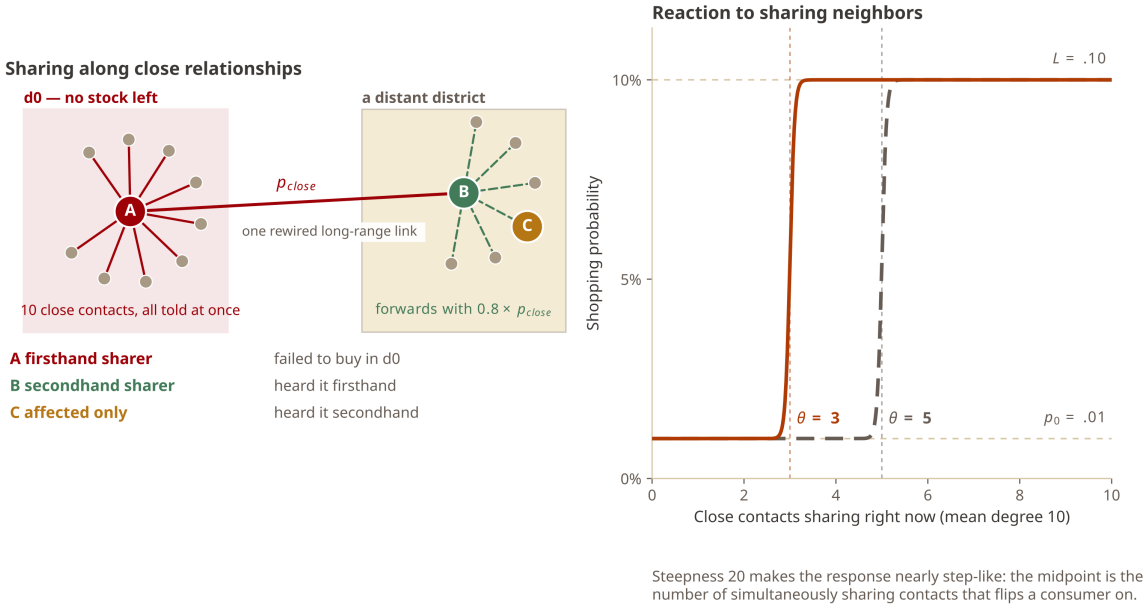


Figure 5: Left: a consumer who fails to buy in the empty district tells all 10 close contacts at once with probability p_{close}, and anyone who hears it can forward it once with probability $0.8 \times p_{close}$. Right: the shopping response at the two reaction midpoints used in the four-treatment summary.

5.1.6 Shortage feedback

Shortage feedback is a switch. When it is on, a failed purchase in any empty district creates new firsthand information. When it is off, only failures in the seeded district 0 do, so all information traces back to the seeded shortage. Comparing the two shows how much the loop in Figure 2 adds beyond the spread of the initial signal.

5.1.7 One simulation step

1. Sharing scheduled in the previous step becomes active. Every consumer counts the contacts who are currently sharing.

2. A consumer who hears the information for the first time may schedule a secondhand share for the next step.

3. The count sets the shopping probability. Consumers who decide to shop are committed and go to their home district first.

4. Each district allocates its remaining stock at random among that step's shoppers. A buyer leaves the market. A consumer who fails witnesses the shortage, may schedule a firsthand share for the next step, and moves to the next district on the route.

5. Sharing that has lasted 10 steps expires.

Each consumer decides at most once whether to share firsthand information and at most once whether to forward secondhand information. The outcome is the *depletion time*: the first step at which 97% of the districts have run out of stock. With 25 districts this means all districts. Runs stop at 150 steps. Lower depletion times mean faster depletion.

5.2 Simulation experiments

All experiments use the settings in Table 4 and change only the parameters listed for each experiment. Consumers are homogeneous: every consumer has the same close-sharing probability and reaction midpoint. Each condition was run with 15 matched seeds (42–56), and every reported value is the mean of these 15 runs. Each experiment was run twice with the same seeds and the same realized networks, once with shortage feedback off and once with it on. The paired difference (on minus off) is negative when feedback makes depletion faster. Initialization, shopping decisions, sharing decisions, stock allocation and network generation each use a separate random stream, so a different number of sharing draws cannot shift the shopping or allocation draws.

Table 4: Settings shared by all network experiments.

Setting	Value
Consumers	2,500
Districts	25 (100 residents each)
Stock per district	50 (abundance .5)
Shopping route	home + 2 districts each side
Initial degree	10
Secondhand sharing	$0.8 \times p_{close}$
Sharing duration	10 steps
Baseline / maximum shop prob.	.01 / .10
Sigmoid steepness G	20
Seeded shortage	district 0
Depletion criterion	97% of districts empty
Maximum steps	150
Seeds	42–56 (15 per condition)

The experiments were:

1. *Sharing only.* p_{close} from 0 to 1 in steps of .1, with a sensitive shopping response ($M = .5$), so that one sharing contact raises the shopping probability from .01 to about .10. Rewiring from 0% to 10% in steps of .5%.

2. *Reaction midpoint only.* M from .5 to 10 in steps of .5, with certain sharing ($p_{close} = 1$). Same rewiring levels.

3. *Midpoint × sharing.* $M \in \{.5, 1, 2, \ldots, 10\}$ crossed with p_{close} from 0 to 1 in steps of .1, at rewiring 0, .5, 1, 2, 3, 4, 6, 8 and 10%.

4. *Four-treatment summary.* A conservative setting ($M = 5$, $p_{close} = .4$), one change in sharing ($p_{close} = .6$), one change in the midpoint ($M = 3$), and both changes together, across rewiring from 0% to 10%.

5. *District granularity.* The four treatments at 25 districts and at 125 districts of 20 residents and 10 units each. Total stock stays at 1,250 units.

Experiments 1–3 contain 52,200 run records. Overlapping conditions were reused, so they required 47,340 unique simulations. All runs reached the depletion criterion within 150 steps (range 12–86 steps). The simulation code is `src/network_rewiring_experiment.py`.

5.3 Results

5.3.1 Baseline without information

With sharing switched off, mean depletion time was 73.5 steps at every rewiring level. Rewiring cannot matter when nothing is transmitted. This value is the reference for all other conditions.

5.3.2 Sharing alone

With a sensitive shopping response, the effect of sharing depended on rewiring (Figure 6). Weak sharing changed little: at $p_{close} = .2$, depletion time went from 72.9 steps on the unrewired ring to 69.3 steps at 10% rewiring. Stronger sharing produced a clear small-world transition. At $p_{close} = .4$, depletion time fell from 72.2 to 28.1 steps across the rewiring range, and at $p_{close} = .6$ from 70.7 to 16.7 steps. At 10% rewiring the means were 14.6 steps for $p_{close} = .8$ and 13.7 steps for $p_{close} = 1$. Most of the acceleration happened within the first few percentage points of rewiring.

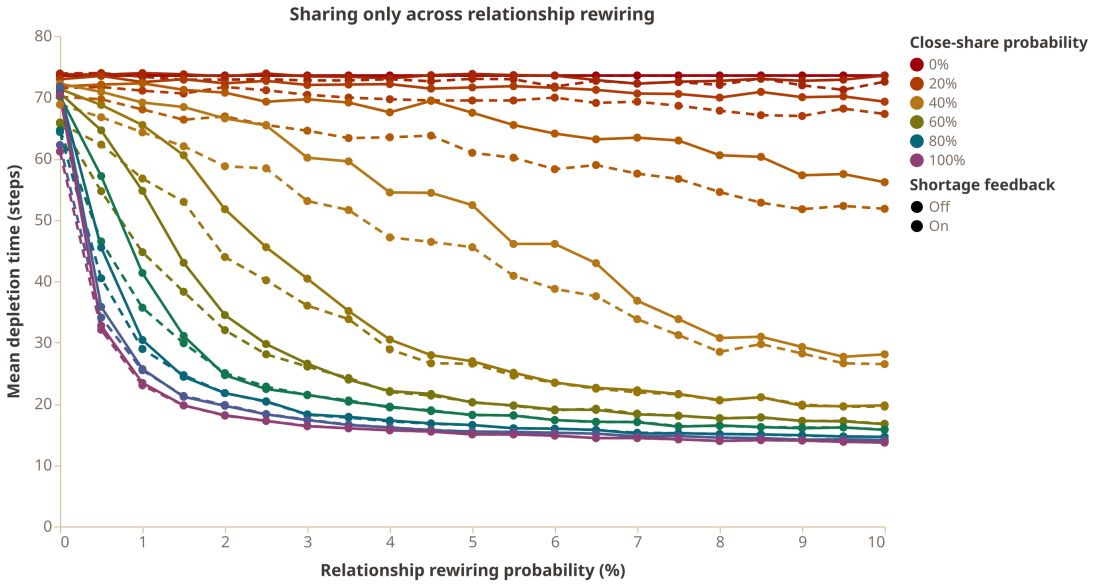


Figure 6: Sharing-only sweep across rewiring ($M = .5$). Solid lines: shortage feedback off; dashed lines: feedback on. Each point is the mean of 15 matched runs.

5.3.3 Reaction midpoint alone

With certain sharing, the midpoint mattered only after rewiring (Figure 7). On the unrewired ring every midpoint stayed near the baseline (70.3–73.5 steps), because information moved slowly along the local ring. For midpoints from .5 to 6, rewiring strongly accelerated depletion: at 1%

rewiring the means ranged from 23.0 to 27.9 steps, and at 10% from 13.7 to 17.1 steps. In this region a few long-range relationships mattered much more than extra sensitivity. Midpoints of 7 and above moved back toward the baseline. At 10% rewiring, the mean was 28.3 steps at $M = 7$, 47.4 at $M = 8$, 61.6 at $M = 9$ and 70.1 at $M = 10$. A threshold close to the number of contacts shielded consumers from the seeded signal even when every witness shared.

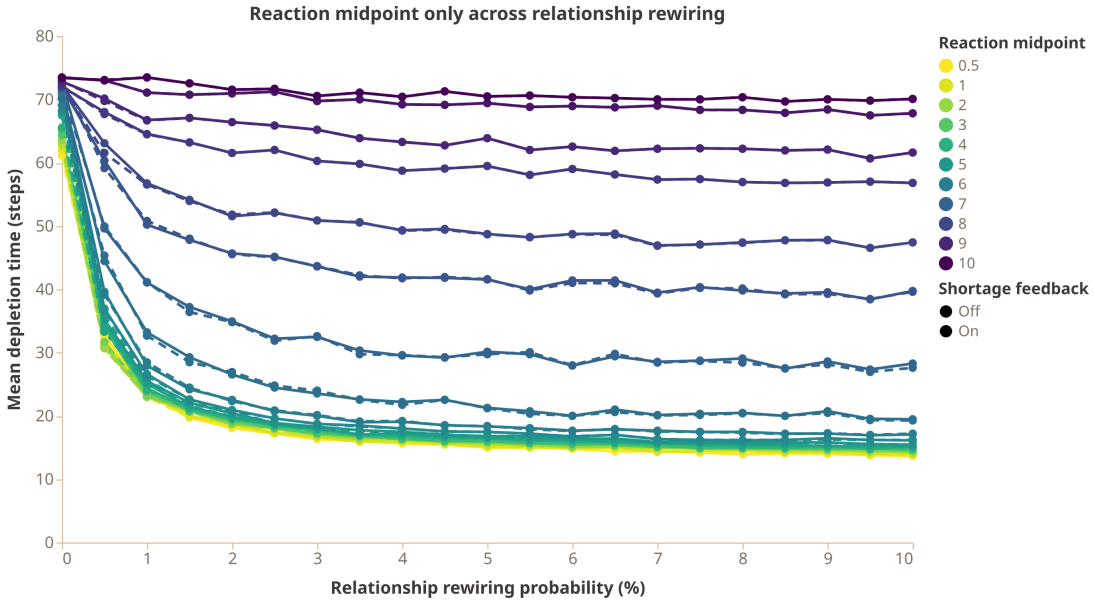


Figure 7: Reaction-midpoint sweep across rewiring ($p_{close} = 1$). Solid lines: shortage feedback off; dashed lines: feedback on.

5.3.4 Sharing and midpoint together

Figure 8 crosses the two parameters. On the unrewired ring the surface was flat: all 121 cell means lay between 70.2 and 74.2 steps, so neither high sharing nor a low midpoint was enough while information stayed local. Rewiring opened a curved boundary between baseline-like and fast depletion. The boundary moved toward lower sharing as rewiring increased, and toward higher sharing as the midpoint increased: less sensitive consumers needed more reliable transmission. At 10% rewiring and $M = .5$, mean depletion time was 69.3 steps at $p_{close} = .2$, 28.1 at .4 and 16.7 at .6. At $M = 5$ the transition came later: 69.9 steps at $p_{close} = .4$, 49.7 at .6, 22.5 at .8 and 15.5 at 1.

The largest structural change happened at very low rewiring. The fastest cell fell from 70.2 steps on the unrewired ring to 31.2 steps at .5% rewiring, 23.0 at 1% and 18.1 at 2%. No cell reached 30 steps or less at .5% rewiring, 13 cells did at 1%, 22 at 2% and 34 at 10%, while the fastest cell improved more slowly, to 13.7 steps at 10%.

5.3.5 Four-treatment summary

Figure 9 shows the joint threshold with four treatments. The conservative setting ($M = 5$, $p_{close} = .4$) was nearly inert: its mean fell only from 73.2 to 69.9 steps across the rewiring range, within one seed-level standard deviation of the 73.5-step baseline. Each single change had a partial effect. At 10% rewiring with feedback off, raising sharing to .6 gave 49.7 steps

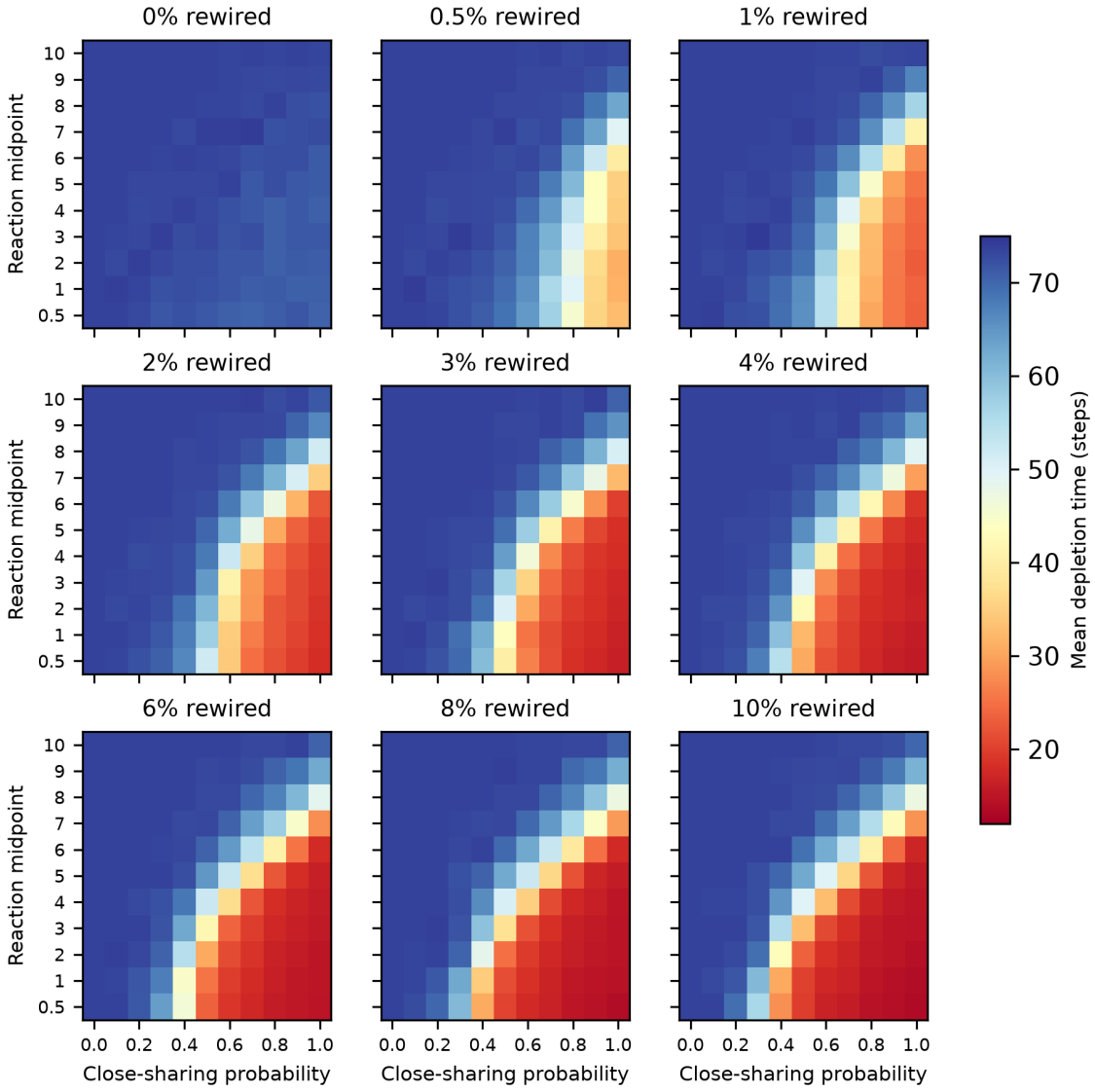


Figure 8: Mean depletion time (steps) across reaction midpoint and close-sharing probability at nine rewiring levels, shortage feedback off. Each cell is the mean of 15 runs. Blue: slow, baseline-like depletion; red: fast depletion.

and lowering the midpoint to 3 gave 55.3 steps, reductions of 20.2 and 14.6 steps. Both changes together gave 20.9 steps, a reduction of 49.0 steps, or 70% faster depletion.

The combination beat the sum of its parts, and the excess was largest where the small-world transition happened. At 2% rewiring, the sharing change and the midpoint change reduced depletion time by 10.7 and 0.9 steps, but both together reduced it by 31.8 steps, an interaction of −20.1 steps. The interaction was −24.0 steps at 4% rewiring and −14.2 steps at 10%, where the single changes had themselves grown. All four treatments stayed near the baseline on the unrewired ring.

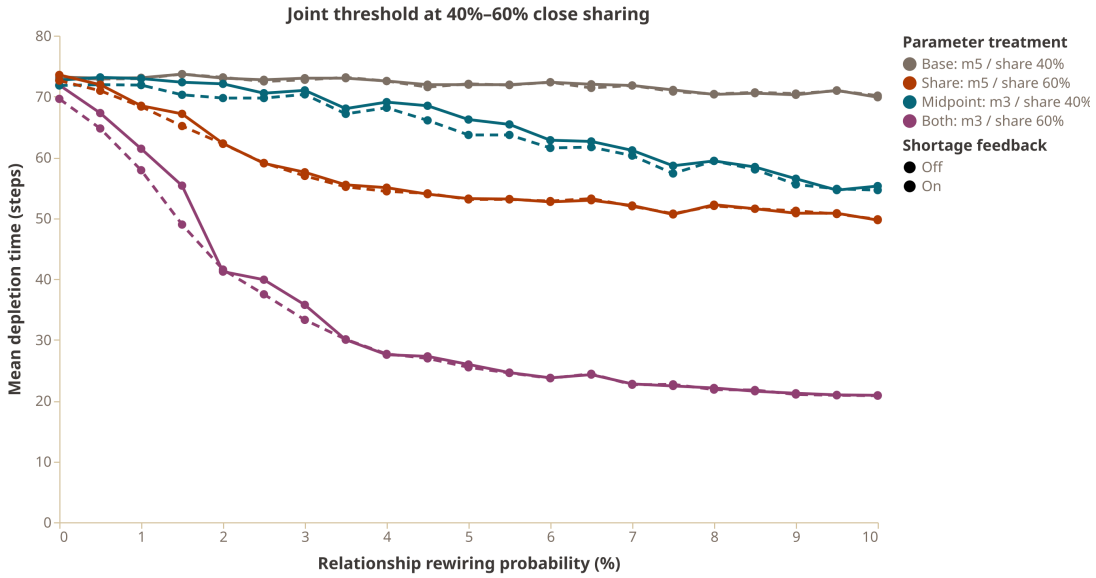


Figure 9: Joint threshold around $M = 5$ and $p_{close} = .4$. Colors: the conservative setting, each single change, and both changes. Solid lines: shortage feedback off; dashed lines: feedback on. Each point is the mean of 15 matched runs.

5.3.6 Shortage feedback

Shortage feedback had little effect at this district size. Across all 16,335 matched pairs of the midpoint × sharing experiment, feedback shortened depletion by 0.64 steps on average (mean absolute difference 0.96 steps). Its effect was largest near the transmission boundary. In the sharing-only experiment, the mean paired difference was 0 at $p_{close} = 0$, −1.99 steps at .2, −4.96 at .3, −4.83 at .4, −1.62 at .6 and −0.49 at 1. Where information is absent, feedback has nothing to amplify; where information already spreads reliably, it adds little. In the four-treatment summary, the feedback-on means at 10% rewiring were 70.1, 49.8, 54.6 and 20.8 steps, almost the same as with feedback off.

Feedback became more important with smaller districts. Feedback can only matter if districts other than the seeded one run out while the process can still react to the information they create, and a 50-unit district runs out late. With 125 districts of 10 units, a second district ran out earlier in the run: in the conservative treatment, at step 23.6 of a 79.3-step run (30% of the way through) instead of step 39.4 of a 73.2-step run (54%). Pooled across treatments and rewiring levels, the mean paired difference grew from −0.44 steps at 25 districts to −1.79 steps at 125 districts. The change was concentrated in the combined treatment (from −0.28 to −3.97 steps)

and at low rewiring, where the seeded signal alone spreads slowly: −7.0 steps on the unrewired ring and −8.4 steps at 2% rewiring, compared with −2.3 and +0.8 steps at 25 districts (Table 5). From 4% rewiring on, the two resolutions converged, because long-range relationships already spread the seeded signal.

Table 5: Mean depletion time (steps) of the combined treatment ($M = 3$, $p_{close} = .6$) with shortage feedback off and on, at two district resolutions. Diff = on minus off.

Rewiring (%)	25 off	25 on	Diff	125 off	125 on	Diff
0	71.5	69.2	-2.3	80.3	73.3	-7.0
2	40.0	40.8	+0.8	66.1	57.7	-8.4
4	26.4	26.6	+0.2	50.0	46.4	-3.6
6	22.8	22.7	-0.1	36.7	35.7	-1.0
8	20.9	20.7	-0.2	36.3	33.3	-3.0
10	20.2	20.1	-0.1	30.8	30.0	-0.8

Changing the number of districts also changes other properties that cannot be held constant: route coverage falls from 20% to 4% of the market, the seeded district shrinks from 100 to 20 residents, fewer lattice links stay inside a district (85% instead of 97%), and the 97% criterion requires 122 of 125 districts rather than all 25. The first two work against faster depletion and the last two work for it, so this comparison shows the direction of the feedback effect, not its size.

5.4 Discussion

The simulation shows that a large aggregate response does not require a large change in any single behavior. Neither more sharing nor more sensitive shopping was enough on its own at the conservative setting. Together they made depletion about 70% faster, far more than the sum of their separate effects. This joint threshold is the main result. A market can sit close to a boundary where two modest changes, each hard to detect at the individual level, turn a slow, local event into a fast, market-wide one.

The network structure decides whether this boundary can be crossed. With close relationships only between neighbors, every combination of sharing and sensitivity stayed near the baseline, because information stayed near the seeded district. A small share of long-range relationships was enough to change this: the first 1–2% of rewired links produced most of the acceleration. Close relationships that reach other districts, such as family members or friends who shop elsewhere, can therefore carry a local shortage to the whole market without any public broadcast. This matters for the survey results, where private information-sharing intention, not public information-sharing intention, was the sharing behavior most consistently tied to subjective prospects.

The shortage-feedback loop, in which new shortages create new information, added little at the default district size and more with small districts. Its role therefore depends on how quickly local shortages appear relative to the spread of the first signal. With large stores, the seeded signal decides the outcome before other stores run out. With small stores, secondary shortages appear early enough to feed the process.

The model has several limitations. The parameters are not estimated from data. The surveys

measured intentions on a 1–5 scale, not probabilities, so they constrain the structure of the model (one sharing propensity, a weaker secondhand channel, sharing with a few close others) rather than its values. Consumers are homogeneous, the network is fixed, stock is not replenished and public SNS posting is not modeled. The shopping response is close to a step function, which makes the midpoint act like a threshold. The granularity comparison changes several properties at once. Finally, depletion time measures how fast the market runs out, not whether a real panic-buying episode is recognized as one.

6 General Discussion

This thesis asked how panic buying can empty a whole market in a few days when the behavior of each consumer changes only a little. Two surveys measured how consumers respond to shortage information, and an agent-based model asked what these responses imply for a market.

6.1 Small individual differences

At the individual level, the effects were consistently small. In Study 1, news about the future supply of rice changed subjective prospects by more than one standard deviation but changed purchase and information-sharing intentions little or not at all. In both studies, participants with more negative subjective prospects had higher purchase intention and higher private information-sharing intention, with correlations of about .18–.20 (Table 3). These correlations were similar across rice, toilet paper and laundry detergent, across the positive and negative prospects conditions, and with and without the crisis context. In Study 1, the correlations did not differ detectably between the news conditions, even though the conditions differed strongly in subjective prospects. The link between subjective prospects and behavior was therefore stable, and it was weak: subjective prospects accounted for about 3–4% of the variance in purchase intention and private information-sharing intention.

The structure of the responses was the most stable result. In both studies, responses separated into purchase intention, private information-sharing intention, and public information-sharing intention, and not by whether the shortage was observed directly on a shelf or indirectly in an SNS post. Public information-sharing intention was low in both studies. Purchase intention and public information-sharing intention were unrelated, and private information-sharing intention was related to both. The preregistered CFA with independent factors failed, but the EFA recovered the same three groups of items in both studies, so the three types of response are better seen as distinct but correlated.

The role of public information-sharing intention was the main inconsistency. Subjective prospects were unrelated to public information-sharing intention for rice ($r = .05$) but related to it for toilet paper and detergent during the oil-supply crisis ($r = .19$). Whatever its size, the association concerns a behavior that few participants said they would perform. The evidence therefore does not support the idea that consumers with negative prospects drive panic buying mainly by broadcasting shortages to strangers. It is more consistent with shortage information spreading through private information sharing among close relationships.

6.2 From individuals to the market

The simulation shows why small individual differences do not imply small aggregate effects. When private sharing and sensitivity to shared information were both set to modest values, the market behaved as if there were no information at all. Raising either one alone had a partial effect. Raising both made depletion about 70% faster, far more than the sum of the separate effects. Most of this acceleration required only a small share of close relationships that reach other districts. In such a market, small shifts in two behaviors, each hard to detect in a survey, can together turn a local shortage into a market-wide one.

This suggests a view of what changes in an emergency. Study 1 found no sign that negative prospects news changes how strongly subjective prospects are linked to behavior. Study 2 found that concern about the crisis added to purchase intention and private information-sharing intention without amplifying the effect of subjective prospects. An emergency may therefore work less by changing how each person responds and more by shifting many people slightly in the same direction at the same time: slightly more negative prospects, slightly higher willingness to share privately with family and friends, and a slightly quicker shopping response when they hear about a shortage. Common sources such as mass media reports could produce such simultaneous shifts [CITE]. Because the market can sit close to a joint threshold, these shifts do not need to be large.

This view also explains why panic buying is rare despite the constant availability of information channels. In the model, the conservative setting stayed near the baseline across the whole range of network structures. The same networks that carry a panic in an emergency carry little in normal times, because too few people share and too few react.

6.3 Limitations

The surveys measured intentions in hypothetical observation contexts, not actual purchases or messages, and both used online panels. Many responses were excluded for failed comprehension or attention checks (46% of consenting responses in Study 1 and 41% of eligible responses in Study 2), so the samples are selected toward attentive participants. In Study 1 the comprehension checks excluded more participants in the positive prospects condition, which weakens the comparison between conditions. The correlations are cross-sectional and do not show that subjective prospects cause purchase or sharing. The simulation is not calibrated: its parameters are chosen to explore the model, and the surveys constrain only its structure. The model assumes homogeneous consumers, a fixed network, no restocking, and no public information sharing.

6.4 Future directions

Three steps would connect the parts of this thesis more closely. First, the distributions of information-sharing and purchase intentions in the surveys could inform heterogeneous consumers in the model, instead of one value for everyone. Second, the number of close contacts who live and shop in other areas could be measured directly, since the simulation shows that this share decides whether a local shortage can spread. Third, observed purchase data or messaging data from a real shortage would show whether the small differences in intentions found here

appear in behavior.

6.5 Conclusion

Individual consumers respond to shortage information in modest and stable ways. They report higher purchase intention and private information-sharing intention when they hold more negative subjective prospects about supply, and they rarely broadcast shortages publicly. The simulation shows that such modest responses are enough for a market-wide shortage when several of them shift together and when close relationships reach beyond the local neighborhood. Large-scale panic buying does not require panicking consumers; it can emerge from small, simultaneous changes in ordinary behavior.

References

Bentall, R. P., Lloyd, A., Bennett, K., McKay, R., Mason, L., Murphy, J., McBride, O., Hartman, T. K., Gibson-Miller, J., Levita, L., Martinez, A. P., Stocks, T. V. A., Butter, S., Vallières, F., Hyland, P., Karatzias, T., & Shevlin, M. (2021). Pandemic buying: Testing a psychological model of over-purchasing and panic buying using data from the United Kingdom and the Republic of Ireland during the early phase of the COVID-19 pandemic. *PLOS ONE*, 16(1), e0246339. https://doi.org/10.1371/journal.pone.0246339

Iizuka, R., Toriumi, F., Nishiguchi, M., Takano, M., & Yoshida, M. (2022). Impact of correcting misinformation on social disruption. *PLOS ONE*, 17(4), e0265734. https://doi.org/10.1371/journal.pone.0265734

Keane, M., & Neal, T. (2021). Consumer panic in the COVID-19 pandemic. *Journal of Econometrics*, 220(1), 86–105. https://doi.org/10.1016/j.jeconom.2020.07.045

Kome kakaku ni sakiyasukan, 10-gatsu mitōshi shisū, jukyū kanwa de, 4-kagetsu-buri teisui-jun [Rice prices expected to fall: October outlook index at a four-month low as supply and demand ease]. (2025, November 11). *Nihon Keizai Shimbun*, 23.

Nakazawa, S. (2025, October 15). Dōtō, "rikutō" saibai ni netsushisen: Reiryō na kikō demo ittei no shūkakuryō kakuho [Eastern Hokkaido turns to upland rice: Stable yields even in a cool climate]. *Asahi Shimbun*, 27.

Ohinata, H. (2026, January 21). "Owatta" ten aogu nōsui kanryō: Shōgeki dēta de jishin sōshitsu, soko ni rinji jōhō [Reiwa no kome sōdō] ["It's over," farm ministry bureaucrat looks up at the sky: Confidence lost after shocking data, then comes the extra earthquake advisory (Reiwa rice crisis)]. *Asahi Shimbun Digital*. https://digital.asahi.com/articles/ASV1J35ZJV1JULFA02WM.html

Shinmai, zōsanka no neagari: 5-kiro 4000-en-dai teichaku, kyōkyū risuku jōtaika [New rice prices rise despite higher production: Over ¥4,000 per 5 kg becomes the norm, and supply risk becomes chronic]. (2025, October 18). *Nihon Keizai Shimbun*, 5.

The Mainichi. (2024, November 2). [Article on the August–September 2024 rice distribution disruptions following the Nankai Trough Earthquake Extra Information advisory; TODO: insert exact headline]. https://mainichi.jp/english/articles/20241102/p2a/00m/0bu/017000c

Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of "small-world" networks. *Nature*, 393(6684), 440–442. https://doi.org/10.1038/30918

Yasuda, T. (2025, October 15). Heiyabu no beisaku, shōrai wa fukanō? Mōsho de tekion koe: Mie-dai kenkyū gurūpu chōsa [Rice farming on the plains impossible in the future? Extreme heat exceeds the optimal temperature: A study by a Mie University research group]. *Asahi Shimbun*, 25.

Zuo, X., & Ohtsubo, Y. (2025). Kaishime sawagi ni okeru jōhō kyōyū keikō to jōhō hannō ikichi no eikyō [Effects of information-sharing tendency and information-reaction threshold in panic buying]. *Proceedings of the 71st Annual Meeting of the Japanese Group Dynamics Association*. https://doi.org/10.2130/jjesp.65-S1 [CHECK]

A Appendix

Placeholder, to be completed. Details moved out of the main text are collected here.

A.1 Behavioral items

Table 6: Behavioral items in Studies 1 and 2 (translated from Japanese). Each item was answered on a 5-point scale from *definitely not* to *definitely*. Target is the factor found in the Study 1 EFA and preregistered for Study 2.

Context	Target	Study 1 (rice)	Study 2 (toilet paper, laundry detergent)
Direct	Purchase	Buy the remaining rice; a few days of rice left at home	Buy the remaining stock; only a few days' worth at home
Direct	Purchase	Same; more than a week of rice left	Same; about half the usual amount at home
Direct	Purchase	Same; more than a month of rice left	Same; the usual amount at home
Direct	Private sharing	Tell housemates (family), best friends or partner	Tell housemates (family), best friends or partner
Direct	Private sharing	Tell acquaintances or friends	Tell acquaintances or friends
Direct	Public sharing	Post to many unspecified people on SNS	Post to many people on SNS
Indirect	Purchase	Go to buy rice right away; a few days of rice left	Go to buy right away; only a few days' worth
Indirect	Purchase	Same; more than a week of rice left	Same; about half the usual amount
Indirect	Purchase	Same; more than a month of rice left	Same; the usual amount
Indirect	Private sharing	Tell housemates (family), best friends or partner	Tell housemates (family), best friends or partner
Indirect	Private sharing	Tell acquaintances or friends	Tell acquaintances or friends
Indirect	Public sharing	Share or post to many unspecified people	Repost or share to many people

A.2 Study 1

A.2.1 Exclusions

Responses were excluded in the order shown in Table 7. Test responses were recorded before the survey was launched on January 7, 2026. We dropped every response whose participant ID or IP address appeared more than once, because we could not tell which copy was valid. The comprehension checks excluded every participant with a wrong answer to any of the four questions of the assigned condition. Among consenting responses, they excluded 100 of 265 in the negative prospects condition and 146 of 265 in the positive prospects condition. We also inspected response duration. The shortest response took 70 seconds; because the survey was short and these participants passed all comprehension checks, no response was excluded for duration.

Table 7: Study 1: responses excluded at each step.

Step	Excluded	Remaining
Raw responses		594
Unfinished	0	594
Test responses before launch	19	575
Participant ID or IP address repeated	6	569
No consent	39	530
Any wrong comprehension answer	246	284

A.2.2 Household questions

Before reading the articles, participants reported their household size, whether they were the main buyer of food in the household, how often per week they cooked rice and ate rice, and how much rice they cooked at a time. [PLACEHOLDER: descriptive statistics, and the condition comparisons and correlations repeated with these variables as controls.]

A.2.3 Behavioral variables

Expected structure. We had designed the items expecting the observation context, direct or indirect observation, to separate the items, and the sharing items to form one factor. Neither happened: purchase items from both contexts loaded on one factor, and sharing split into private and public information-sharing intentions.

Promax rotation. The promax solution gave the same three factors as the varimax solution. The factor correlations were .56 between purchase intention and private information-sharing intention, .18 between private and public information-sharing intention, and -.01 between purchase intention and public information-sharing intention. Table 8 shows the promax loadings.

Regression factor scores. We repeated the condition comparisons and the correlations with regression factor scores from the varimax solution instead of item means. Private information-sharing intention was again slightly higher in the negative prospects condition ($t[257] = 2.15$, $p = .032$, $d = 0.26$); purchase intention ($p = .760$) and public information-sharing intention ($p = .977$) did not differ. Subjective prospects correlated with purchase intention ($r = .16$,

Table 8: Promax-rotated loadings of the twelve behavioral items in Study 1 (rice, $N = 284$). Minimum residual extraction; loadings $\geq .40$ in bold.

Item	Context	Purchase	Private	Public
Buy: a few days left	Direct	**.48**	.19	-.30
Buy: a week left	Direct	**.80**	.03	-.21
Buy: a month left	Direct	**.82**	-.14	.15
Go to buy: a few days	Indirect	**.58**	.21	-.08
Go to buy: a week	Indirect	**.83**	.00	.03
Go to buy: a month	Indirect	**.82**	-.15	.29
Tell housemates/close	Direct	-.06	**.87**	-.14
Tell acquaintances	Direct	-.13	**.82**	.17
Tell housemates/close	Indirect	.15	**.67**	.05
Tell acquaintances	Indirect	.08	**.65**	.29
Post to many on SNS	Direct	-.01	.11	**.79**
Share/repost on SNS	Indirect	.04	.06	**.85**

$p = .006$) and private information-sharing intention ($r = .15$, $p = .014$) but not with public information-sharing intention ($r = .03$, $p = .589$).

A.2.4 Effect of the news condition

Table 9: Study 1: subjective prospects and behavioral intentions by news condition, with Welch's t-tests (negative minus positive). Values are M (SD).

Measure	Negative ($n = 165$)	Positive ($n = 119$)	t	df	p	
Subjective prospects	4.05 (0.66)	3.12 (0.76)	10.75	231	$< .001$	1.3
Purchase intention	2.79 (0.83)	2.73 (0.78)	0.64	263	.521	0.0
Private information-sharing intention	2.91 (0.88)	2.70 (0.85)	2.02	258	.044	0.2
Public information-sharing intention	1.72 (0.85)	1.70 (0.75)	0.18	270	.853	0.0

A.2.5 Correlations among the variables

Table 10: Study 1: means, standard deviations and Pearson correlations of subjective prospects and the behavioral intentions, pooled over conditions ($N = 284$). 95% confidence intervals in brackets. $* p < .05$, $** p < .01$, $*** p < .001$.

Variable	M	SD	1	2	3
1. Subjective prospects	3.66	0.84			
2. Purchase intention	2.77	0.81	.18** [.07, .29]		
3. Private information-sharing intention	2.83	0.87	.18** [.06, .29]	.49*** [.39, .57]	
4. Public information-sharing intention	1.71	0.81	.05 [-.07, .16]	.05 [-.07, .16]	.30*** [.19, .41]

Table 11: Study 1: correlations between subjective prospects and the behavioral intentions within each news condition, compared with Fisher's r-to-z test.

Outcome	r (negative, $n = 165$)	r (positive, $n = 119$)	z	p
Purchase intention	.26	.12	1.15	.249
Private information-sharing intention	.18	.08	0.84	.398
Public information-sharing intention	.05	.05	0.00	$> .999$

A.2.6 Correlations within each condition

A.2.7 Sharing after direct and indirect observation

Averaging the two items about private information-sharing intention within each observation context, participants were more willing to share a shortage they had observed directly ($M = 2.96$, $SD = 0.95$) than one they had observed indirectly in an SNS post ($M = 2.69$, $SD = 0.95$; $t[283] = 5.85$, $p < .001$, $d_z = 0.35$). The two composites were strongly correlated ($r = .68$). After rescaling both means to 0–1, sharing after indirect observation was 87% of sharing after direct observation. The difference did not depend on the condition: it was 0.32 in the negative prospects condition and 0.19 in the positive prospects condition (Welch's $t[262] = 1.43$, $p = .155$, $d = 0.17$).

A.3 Study 2

A.3.1 Exclusions

During data collection, Qualtrics did not count responses with a wrong answer in the attention checks toward the planned sample of 500. As preregistered, we closed the survey once the counted responses reached 500, and participants who had already started were allowed to finish, which gave 1,028 raw responses. Responses were excluded by the preregistered rules, in the order shown in Table 12. Six attention checks were embedded in the observation context blocks and at the end of the survey: three asked whether the participant had looked at the image, two asked for a specified answer, and one asked whether the participant had read the questions carefully. A last check for a repeated participant ID or IP address, not in the preregistration, excluded no response.

Table 12: Study 2: responses excluded at each step.

Step	Excluded	Remaining
Raw responses		1028
Unfinished	0	1028
No consent	80	948
Age outside 30–60	44	904
Duration under 60 s	3	901
Any failed attention check	365	536
Missing analysis variables	0	536
Participant ID or IP address repeated	0	536

A.3.2 Robustness checks

Number of factors. As in Study 1, we kept the three factors with eigenvalues greater than 1; the fourth and fifth eigenvalues of the item correlation matrix were 0.83 and 0.69.

Promax rotation. Every item loaded highest on its preregistered factor in both the varimax solution (purchase intention .67–.81, private information-sharing intention .72–.78, public information-sharing intention .76 and .90) and the promax solution. The promax factor correlations were .48 between purchase intention and private information-sharing intention, .44 between private and public information-sharing intention, and .07 between purchase intention and public information-sharing intention. [PLACEHOLDER: promax loading table.]

A.3.3 Exploratory analyses

Commodity. Subjective prospects were more negative for laundry detergent ($M = 3.06$, $SD = 0.60$) than for toilet paper ($M = 2.85$, $SD = 0.70$; $t[504] = 3.71$, $p < .001$, $d = 0.32$). Purchase intention was higher for toilet paper ($M = 3.56$, $SD = 0.80$) than for laundry detergent ($M = 3.33$, $SD = 0.87$; $t[534] = 3.23$, $p = .001$, $d = 0.28$), while private information-sharing intention, public information-sharing intention, and crisis concern did not differ ($p \geq .114$). Because subjective prospects differed between commodities, the preregistered exploratory plan called for commodity-specific correlations. They were similar for both commodities: for toilet paper ($n = 257$), $r = .21$ (purchase intention), .19 (private information-sharing intention), and .16 (public information-sharing intention); for laundry detergent ($n = 279$), $r = .20$, .21, and .24 (all $p \leq .009$).

Crisis context. Three exploratory items asked how concerned participants were about the oil-supply situation, how much it would affect their daily shopping, and how long it would last (5-point scales; $M = 3.91$, $SD = 0.84$, $\alpha = .67$). Concern about the crisis was related to subjective prospects about the commodity ($r = .17$, $p < .001$), to purchase intention ($r = .26$, $p < .001$), and to private information-sharing intention ($r = .18$, $p < .001$), but not to public information-sharing intention ($r = -.05$, $p = .255$). In regressions with commodity, subjective prospects, crisis context and their interaction (predictors mean-centered), both subjective prospects and crisis context predicted purchase intention ($b = 0.21$ and 0.24, both $p < .001$) and private information-sharing intention ($b = 0.25$ and 0.18, both $p < .001$). For public information-sharing intention, only subjective prospects did ($b = 0.27$, $p < .001$; crisis $b = -0.08$, $p = .061$). The interaction was not significant for any outcome (all $p \geq .222$).

A.3.4 Sharing after direct and indirect observation

Averaging the two items about private information-sharing intention within each observation context, participants were more willing to share a shortage they had observed directly ($M = 3.01$, $SD = 1.00$) than one they had observed indirectly in an SNS post ($M = 2.87$, $SD = 1.01$; $t[535] = 4.47$, $p < .001$, $d_z = 0.19$). The two composites were strongly correlated ($r = .75$). After rescaling both means to 0–1, sharing after indirect observation was 93% of sharing after direct observation. [PLACEHOLDER: test whether the difference between direct and indirect observation depends on subjective prospects.]